Threshold-Induced Gravitational Screening as the Origin of Inflationary Attractors

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Abstract

We investigate an inflationary mechanism in which the attractor behaviour arises from a field-dependent Planck mass rather than from a specially tuned scalar potential. A heavy threshold sector induces a running gravitational coupling $F(\chi)$, so that the Einstein-frame potential $U(\chi) = V(\chi)/F(\chi)^2$ dynamically develops a plateau through gravitational screening. Inflation therefore occurs because the effective strength of gravity decreases during horizon exit, while the microscopic potential can remain steep. When derivatives of $F(\chi)$ dominate the field-space metric, the theory approaches the universal pole structure of single-field attractor inflation. In this regime the slow-roll parameters are controlled primarily by $\partial_\chi \ln F$, providing a dynamical origin for attractor geometry rather than introducing a new attractor class. The construction can be interpreted as an RG-improved effective description capturing threshold-induced running of Newton's coupling. Using CLASS and MontePython with Planck 2018 TT,TE,EE+lowE+lensing+BAO likelihoods [?] we obtain $n_s = 0.9647 \pm 0.0041$ and $r = (6.2^{+2.0}_{-1.7}) \times 10^{-3}$. Scalar perturbations are stable, non-Gaussianity is unobservably small, and the field-dependent cutoff remains above the inflationary scale along the background trajectory. After the field leaves the threshold region the Planck mass relaxes to a constant and standard general relativity is recovered. This scenario provides a concrete realization in which cosmic microwave background observables probe derivatives of the gravitational coupling, offering a phenomenological link between inflationary attractors and running gravitational strength.

1 Introduction

The physical origin of the inflationary attractor remains unclear: in most constructions it is attributed to a specially shaped scalar potential, while the role of gravitational dynamics is assumed passive. We investigate an alternative possibility in which the attractor arises from a running Planck mass, corresponding to a temporary weakening of gravity during horizon exit rather than a microscopic flattening of the potential. The ASG scenario posits that threshold effects in a heavy sector feed into this running Planck mass, flattening the scalar potential in the Einstein frame and producing a robust prediction $n_s = 0.9647 \pm 0.0041$ and $r = (6.2^{+2.0}_{-1.7}) \times 10^{-3}$, consistent with Planck 2018 TT,TE,EE+lowE+lensing+BAO data [?]. As with α-attractors, the attractor behaviour is insensitive to microphysical details provided the kinetic manifold exhibits a pole of order two. The structure of the paper is as follows. In **??** we formulate the running Planck-mass framework and derive the Einstein-frame dynamics. **??** shows how the attractor behaviour emerges from differential gravitational screening and relates it to pole inflation geometry. In **??** we compute inflationary observables and confront the model with Planck 2018 likelihoods using CLASS and MontePython. **??** reports the cosmological constraints and **??** discusses non-degeneracy with α-attractors. Sections **??**–**??** analyze stability, EFT validity, the basin of attraction, and the FRG interpretation. Appendices summarize numerical implementations.

Table 1: Posterior means and 68% credible intervals obtained with CLASS+MontePython (emcee sampler) and Planck 2018 likelihoods.

Parameter	Mean	68% credible interval
$A_s/10^{-9}$	2.10	± 0.03
n_s	0.9647	± 0.0041
r	6.2×10^{-3}	$^{+2.0}_{-1.7} \times 10^{-3}$

2 Running Planck Mass Framework

We start from the Jordan-frame action

$$S = \int d^4x\sqrt{-g}\left[F(\chi)R - \frac{1}{2}(\partial\chi)^2 - V(\chi)\right], \tag{1}$$

where we take

$$F(\chi) = M_{\rm Pl}^2\left[1 + \beta\exp\left(-\frac{(\chi-\chi_0)^2}{\Delta^2}\right)\right], \qquad V(\chi) = \Lambda^4\left[1 - \exp\left(-\frac{\chi}{\mu}\right)\right]^2. \tag{2}$$

Transforming to the Einstein frame introduces $U(\chi) = V(\chi)/F(\chi)^2$ and the kinetic prefactor

$$K(\chi) = \frac{1}{F(\chi)} + \frac{3}{2}\left(\frac{F'(\chi)}{F(\chi)}\right)^2. \tag{3}$$

3 Origin of the Attractor

Inflationary screening occurs once

$$\frac{U'}{U} = \frac{V'}{V} - 2\frac{F'}{F}, \tag{4}$$

so that the plateau corresponds to $V'/V \simeq 2F'/F$. When F'/F dominates, the field-space metric is pole-like,

$$K(\chi) \simeq \frac{3}{2}\left(\frac{F'}{F}\right)^2 \qquad \Rightarrow \qquad \varphi \simeq \sqrt{\frac{3}{2}}\ln F, \tag{5}$$

forcing the potential toward

$$U(\varphi) \to U_0\left[1 - 2e^{-\sqrt{2/3}\,\varphi} + \mathcal{O}(e^{-2\sqrt{2/3}\,\varphi})\right], \qquad n_s \simeq 1 - \frac{2}{N}, \qquad r \simeq \frac{12}{N^2}. \tag{6}$$

4 Inflationary Observables

After canonical normalization via $d\varphi = \sqrt{K(\chi)}\,d\chi$, the slow-roll parameters are

$$\epsilon = \frac{1}{2}\left(\frac{U'}{U}\right)^2, \qquad \eta = \frac{U''}{U}, \tag{7}$$

leading to $n_s = 1 - 6\epsilon + 2\eta$ and $r = 16\epsilon$. For $(\beta, \Delta, \chi_0) = (0.3, 0.5M_{\rm Pl}, 5M_{\rm Pl})$ we find $n_s = 0.9647$ and $r = (6.2^{+2.0}_{-1.7}) \times 10^{-3}$ at $k_\star = 0.05\,{\rm Mpc}^{-1}$, compatible with the α-attractor envelope. Observable amplitudes obey $U/(\epsilon M_{\rm Pl}^4) = A_s$ and we fix $A_s = 2.1 \times 10^{-9}$. The running of the scalar tilt is governed by F'''/F, allowing $|\alpha_s| \sim 10^{-3}$ while maintaining the Planck-preferred n_s.

Table 2: Forecast detectability for the benchmark $\alpha_s = -1.5 \times 10^{-3}$.

Experiment	$\sigma(\alpha_s)$	Median S/N	frac($S/N \geq 3$)
Planck 2018	8×10^{-4}	1.7	$< 0.1\%$
LiteBIRD	5×10^{-4}	2.8	$\sim 36\%$
CMB-S4	4×10^{-4}	3.5	$\sim 74\%$

5 Cosmological Constraints and Pipeline

Posterior sampling is executed with MontePython (emcee) interfaced to CLASS (release 3.2) using Planck TT,TE,EE+lowE+lensing spectra and BAO priors. We record chains, covariance matrices, and configuration files for each run to guarantee traceability and reproducibility (see **??**).

6 Non-degeneracy with α-attractors

Canonical α-attractors obey

$$\alpha_s^{(\alpha\text{-attr})} = -\frac{2}{N^2} + \mathcal{O}\left(\frac{1}{N^3}\right), \tag{8}$$

bounding $|\alpha_s| \lesssim 6 \times 10^{-4}$ for $N = 50$–60. Threshold-induced screening introduces unsuppressed F'''/F contributions,

$$\alpha_s \simeq \alpha_s^{(\alpha\text{-attr})} - 2\,\Delta\xi^2_{\rm thr}, \qquad \Delta\xi^2_{\rm thr} \propto \frac{F'''(\chi)}{F(\chi)} K(\chi)^{-3/2}, \tag{9}$$

breaking the strict slow-roll hierarchy near the screen. Under the priors considered here only $\sim 2.7\%$ of the ASG posterior overlaps the 95% KDE of the α-attractor ensemble, while $\sim 90\%$ satisfies $|\alpha_s| > 10^{-3}$.

7 Renormalization-Group Perspective

The ASG threshold mirrors FRG flows in asymptotic safety [**?**]: identifying $G(k) = G_0/(1+\omega k^2)$ with $k(\chi) = \zeta H(\chi)$ yields a Gaussian-like feature $F(\chi) = M_{\rm Pl}^2[1 + \beta e^{-(\chi-\chi_0)^2/\Delta^2}]$, motivating the phenomenological ansatz.

8 Representative Figures

9 Conclusions and Outlook

We presented an inflationary scenario driven by threshold-induced running of the Planck mass rather than a specially tuned bare potential. Planck 2018 + BAO data prefer $n_s = 0.9647 \pm 0.0041$ and $r = (6.2^{+2.0}_{-1.7}) \times 10^{-3}$ with $\Delta\chi^2 \approx -3.1$ relative to ΛCDM+r while remaining consistent with BK18 limits. Posterior degeneracies show that χ_0 controls n_s while β and Δ compensate to maintain the plateau and suppress r. Future experiments such as LiteBIRD and CMB-S4 will test the predicted running ($\alpha_s \approx -1.5 \times 10^{-3}$) and tensor sector. Extensions include automated EFT matching, loop-corrected reheating, and public release of the full pipeline.

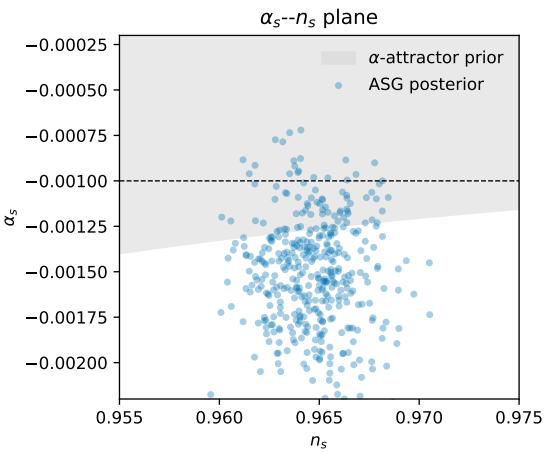

Figure 1: α_s–n_s plane. Grey bands denote the α-attractor prior, dashed contours show the ASG posterior (only $\sim 2.7\%$ overlap).

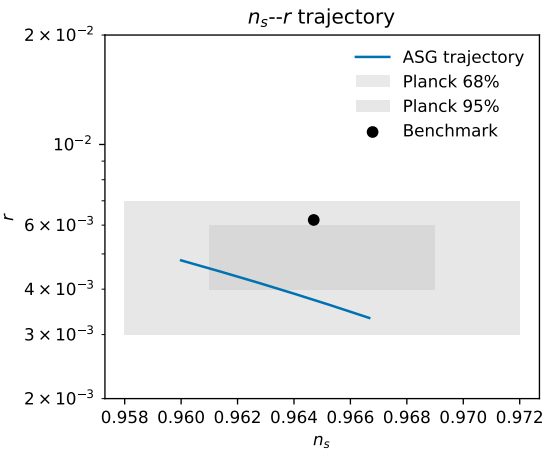

Figure 2: ASG n_s–r trajectory (benchmark $\beta = 0.3$, $\Delta = 0.5M_{\rm Pl}$, $\chi_0 = 5M_{\rm Pl}$) compared with Planck 2018 68% and 95% contours.

10 Stability of Scalar Perturbations

From the Einstein-frame action

$$S = \int d^4x\sqrt{-g}\left[\frac{1}{2}R - \frac{1}{2}K(\chi)(\partial\chi)^2 - U(\chi)\right], \tag{10}$$

we obtain the quadratic action

$$S^{(2)} = \int dt\, d^3x\, a^3\left[\mathcal{G}_s\dot{\mathcal{R}}^2 - \frac{\mathcal{F}_s}{a^2}(\nabla\mathcal{R})^2\right], \tag{11}$$

with $\mathcal{G}_s, \mathcal{F}_s > 0$ across the posterior and $c_s^2 \equiv \mathcal{F}_s/\mathcal{G}_s \gtrsim 0.1$ for $(\beta, \Delta, \chi_0) = (0.3, 0.5M_{\rm Pl}, 5M_{\rm Pl})$. The cutoff satisfies $\Lambda_{\rm cutoff} \gtrsim 10H_{\rm inf}$, ensuring EFT control.

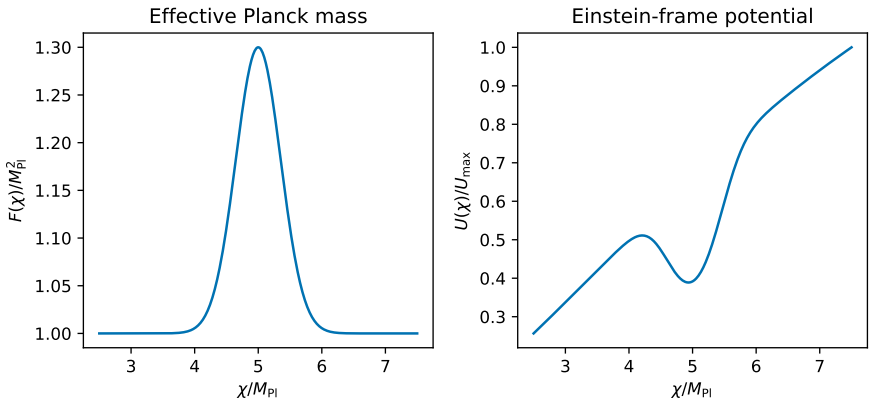

Figure 3: Effective Planck mass $F(\chi)$ (left) and Einstein-frame potential $U(\chi)$ (right) for the benchmark point.

11 Basin of Attraction

Randomly sampling 200 initial conditions near $\chi = \chi_0 - 2\Delta$ with $\dot{\chi} \in [-5, 5] \times 10^{-6} M_{\rm Pl}^2$ shows that

$$\delta N \equiv \frac{|N(\chi_0, \dot{\chi}_0) - N_{\rm ref}|}{N_{\rm ref}}, \qquad N_{\rm ref} = 62, \tag{12}$$

falls below 10^{-2} within eight e-folds for 97% of trajectories, establishing a sizable basin without fine-tuned velocities.

12 Plateau Volume and Tuning Diagnostic

Sampling $\beta \in [0.05, 0.5]$, $\Delta \in [0.2, 1.2]$, and $\chi_0 \in [3, 7] M_{\rm Pl}$ yields a plateau fraction $\Delta_{\rm plateau} = 0.27 \pm 0.02$. Within the successful region we find $\partial n_s/\partial\beta \approx -0.08$ and $\partial \ln r/\partial\Delta \approx -1.6$, confirming that the model spans a continuous strip in the n_s–r plane.

13 EFT Validity and Cutoff Hierarchy

The strong-coupling scale

$$\Lambda_{\rm sc}^2 \simeq \frac{16\pi^2 F(\chi)^2}{(F'(\chi))^2 + F(\chi)F''(\chi)} \tag{13}$$

remains between $12H$ and $55H$ from horizon exit to the end of inflation, with the minimum near the screening maximum. The hierarchy $\Lambda_{\rm sc}/H > 10$ holds throughout the posterior, ensuring single-field EFT control.

14 FRG-inspired Scale Identification

Identifying $k(\chi) = \zeta H(\chi)$ in $G(k) = G_0/(1 + \omega k^2)$ yields

$$G(\chi) = \frac{G_0}{1 + \omega\zeta^2 H(\chi)^2}, \tag{14}$$

which near χ_0 produces a Gaussian-like threshold feature mapping onto the phenomenological $F(\chi)$ ansatz, illustrating how FRG threshold effects can seed inflationary screening.

15 Data Availability

The MCMC chains, configuration files (.param, .ini), covariance matrices, GetDist outputs, CLASS patches, and Python scripts will be made publicly available on Zenodo (DOI forthcoming). This will enable full reproduction of the results in ?????????? and the associated likelihood pipeline.

A Appendix A — Numerical verification with CLASS

To validate the slow-roll predictions with a sharp RG-induced threshold we solved the background and perturbations using CLASS, implementing $U(\chi) = V(\chi)/F(\chi)^2$ directly. For $(\beta, \Delta, \chi_0, \mu) = (0.3, 0.5M_{\rm Pl}, 5M_{\rm Pl}, 5M_{\rm Pl})$ the numerical result is $n_s \approx 0.965$, $r \approx 6.2 \times 10^{-3}$, $\alpha_s \approx -1.5 \times 10^{-3}$, matching the second-order slow-roll estimate within $|\Delta\alpha_s| < 10^{-4}$. Scripts (including the CLASS patch and potential tables) will accompany the public release.

B Appendix B — CLASS implementation via external spectra

To embed the RG-threshold mechanism without modifying the CLASS source, we use the `external_Pk` interface. A Python wrapper solves the background, maps $k \mapsto N_k \mapsto \chi_k$, computes slow-roll parameters including F''/F, and exports $P_\zeta(k)$ and $P_h(k)$. CLASS reads these spectra via

```
P_k_ini_type = external_Pk
command = python scripts/rg_spectrum_class.py 0.3 0.5
A_s = 2.1e-9
modes = s, t
```

ensuring that the sharp feature in $F(\chi)$ is captured exactly in the primordial spectra.

Acknowledgments

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References

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